1. **Software is defined as:**

i) Only the compiled code

ii) Programs, documentation, and operating procedures

iii) Hardware instructions

iv) Data structures only

Correct Answer: ii) Programs, documentation, and operating procedures

2. **The "Software Crisis" was characterized by:**

i) Hardware becoming too expensive

ii) Projects running over budget and behind schedule

iii) Lack of internet connectivity

iv) High energy consumption

Correct Answer: ii) Projects running over budget and behind schedule

3. **Software Engineering is a _____ discipline.**

i) Purely theoretical

ii) Systematic, disciplined, and quantifiable

iii) Random and creative

iv) Hardware-centric

Correct Answer: ii) Systematic, disciplined, and quantifiable

4. **Which is not a characteristic of software?**

i) Software is developed or engineered

ii) Software doesn’t "wear out"

iii) Software is custom-built, not manufactured

iv) Software is a physical entity

Correct Answer: iv) Software is a physical entity

5. **Which of the following is an umbrella activity in software engineering?**

i) Coding

ii) Requirements gathering

iii) Software quality assurance

iv) Design

Correct Answer: iii) Software quality assurance

6. **The Waterfall Model is best suited for:**

i) Projects with changing requirements

ii) Small projects with well-defined requirements

iii) High-risk projects

iv) Complex web applications

Correct Answer: ii) Small projects with well-defined requirements

7. **Which model follows an iterative approach and focuses on risk analysis?**

i) Waterfall Model

ii) Spiral Model

iii) V-Model

iv) RAD Model

Correct Answer: ii) Spiral Model

8. **The V-Model emphasizes:**

i) Rapid prototyping

ii) Coding efficiency

iii) Relationship between testing and development phases

iv) Maintenance costs

Correct Answer: iii) Relationship between testing and development phases

9. **In the Incremental Model, the first increment is usually the:**

i) Core product

ii) User interface

iii) Final product

iv) Documentation

Correct Answer: i) Core product

10. **Prototyping is used to:**

i) Finish the project faster

ii) Help clarify user requirements

iii) Replace the final system

iv) Minimize coding effort

Correct Answer: ii) Help clarify user requirements

11. **The Manifesto for Agile Software Development values:**

i) Processes and tools over individuals

ii) Comprehensive documentation over software

iii) Responding to change over following a plan

iv) Contract negotiation over collaboration

Correct Answer: iii) Responding to change over following a plan

12. **In Scrum, who is responsible for prioritizing the Product Backlog?**

i) Scrum Master

ii) Development Team

iii) Product Owner

iv) Stakeholders

Correct Answer: iii) Product Owner

13. **A "Sprint" in Scrum usually lasts for:**

i) 1 day

ii) 1 to 4 weeks

iii) 6 months

iv) 1 year

Correct Answer: ii) 1 to 4 weeks

14. **Which Agile method uses "User Stories" and "Pair Programming"?**

i) Kanban

ii) XP (Extreme Programming)

iii) Lean

iv) Waterfall

Correct Answer: ii) XP (Extreme Programming)

15. **The Daily Stand-up meeting is intended to:**

i) Assign new tasks

ii) Synchronize activities and report impediments

iii) Demo the product to the client

iv) Conduct performance reviews

Correct Answer: ii) Synchronize activities and report impediments

16. **Requirement Engineering includes which of the following steps?**

i) Elicitation, Analysis, Specification, Validation

ii) Coding, Testing, Maintenance

iii) Design, Implementation, Security

iv) Risk Management and Deployment

Correct Answer: i) Elicitation, Analysis, Specification, Validation

17. **What is an SRS?**

i) Software Resource System

ii) Software Requirements Specification

iii) System Risk Solution

iv) Software Requirement Schedule

Correct Answer: ii) Software Requirements Specification

18. **Non-functional requirements refer to:**

i) What the system does

ii) System constraints and quality attributes (performance, security)

iii) User login details

iv) Database tables

Correct Answer: ii) System constraints and quality attributes (performance, security)

19. **Ambiguity in requirements means:**

i) Requirements are too long

ii) Requirements can be interpreted in multiple ways

iii) Requirements are very clear

iv) Requirements are missing

Correct Answer: ii) Requirements can be interpreted in multiple ways

20. **Which tool is often used to model functional requirements?**

i) Class Diagram

ii) Use Case Diagram

iii) Deployment Diagram

iv) Gantt Chart

Correct Answer: ii) Use Case Diagram

21. **The "Project Triangle" consists of:**

i) Code, Test, Deploy

ii) Time, Cost, Scope

iii) Manager, Developer, Client

iv) Risk, Quality, Value

Correct Answer: ii) Time, Cost, Scope

22. **A Gantt Chart is used for:**

i) Managing code versions

ii) Project scheduling and tracking

iii) Writing requirements

iv) Designing UI

Correct Answer: ii) Project scheduling and tracking

23. **COCOMO stands for:**

i) Common Cost Model

ii) Constructive Cost Model

iii) Computer Costing Mode

iv) Conceptual Coding Model

Correct Answer: ii) Constructive Cost Model

24. **The "Critical Path" in a project network refers to:**

i) The easiest tasks

ii) The longest path of planned activities to the end of the project

iii) The path with the most bugs

iv) The documentation phase

Correct Answer: ii) The longest path of planned activities to the end of the project

25. **WBS stands for:**

i) Work Break Schedule

ii) Work Breakdown Structure

iii) Weekly Business Summary

iv) Website Building System

Correct Answer: ii) Work Breakdown Structure

26. **Which of the following is a "Good Design" principle?**

i) High Coupling, Low Cohesion

ii) Low Coupling, High Cohesion

iii) High Complexity

iv) Large monolithic modules

Correct Answer: ii) Low Coupling, High Cohesion

27. **Cohesion is a measure of:**

i) Interdependence between modules

ii) Functional strength within a module

iii) Number of lines of code

iv) Project cost

Correct Answer: ii) Functional strength within a module

28. **Coupling is a measure of:**

i) Strength of relationship between modules

ii) Comments in the code

iii) Speed of execution

iv) Number of developers

Correct Answer: i) Strength of relationship between modules

29. **Abstraction helps in:**

i) Increasing complexity

ii) Hiding unnecessary details

iii) Writing more code

iv) Slowing down the system

Correct Answer: ii) Hiding unnecessary details

30. **Modularity refers to:**

i) Building software as a single large block

ii) Dividing a system into smaller, independent parts

iii) Using only one programming language

iv) Skipping the design phase

Correct Answer: ii) Dividing a system into smaller, independent parts

31. **The "Golden Rule" of UI design includes:**

i) Make the UI complex to look professional

ii) Place the user in control

iii) Use as many colors as possible

iv) Limit user interaction

Correct Answer: ii) Place the user in control

32. **A "User Persona"**

i) A piece of hardware

ii) A fictional representation of a target user

iii) A database admin

iv) A type of computer virus

Correct Answer: ii) A fictional representation of a target user

33. **Consistency in UI design means:**

i) Elements should look and behave the same across the application

ii) Every page should look different

iii) Using random fonts

iv) Changing button locations frequently

Correct Answer: i) Elements should look and behave the same across the application

34. **The "F-pattern" in UI design refers to:**

i) A coding style

ii) The way users often scan web pages

iii) A type of error

iv) Keyboard shortcuts

Correct Answer: ii) The way users often scan web pages

35. **Affordance in UI design means:**

i) The cost of the UI

ii) A property in which the physical characteristics of an object influence its function

iii) The speed of the UI

iv) The number of images used

Correct Answer: ii) A property in which the physical characteristics of an object influence its function

36. **The Client-Server architecture is a type of:**

i) Monolithic architecture

ii) Distributed architecture

iii) Sequential architecture

iv) Local architecture

Correct Answer: ii) Distributed architecture

37. **MVC stands for:**

i) Model View Controller

ii) Main Video Codec

iii) Module Variable Class

iv) Multi-View Component

Correct Answer: i) Model View Controller

38. **In MVC, the "Model" is responsible for:**

i) The user interface

ii) Data and business logic

iii) Handling user input

iv) Routing

Correct Answer: ii) Data and business logic

39. **Layered architecture helps in:**

i) Making the system slower

ii) Separation of concerns

iii) Combining all logic in one file

iv) Reducing security

Correct Answer: ii) Separation of concerns

40. **Microservices architecture involves:**

i) One giant database

ii) Small, independent services communicating over a network

iii) A single executable file

iv) No documentation

Correct Answer: ii) Small, independent services communicating over a network

41. **Refactoring is the process of:**

i) Adding new features

ii) Improving internal code structure without changing external behavior

iii) Deleting the whole project

iv) Changing the project manager

Correct Answer: ii) Improving internal code structure without changing external behavior

42. **Code reviews are conducted to:**

i) Embarrass the developer

ii) Find bugs and ensure coding standards

iii) Increase the cost

iv) Stop the project

Correct Answer: ii) Find bugs and ensure coding standards

43. **A "Technical Debt" occurs when:**

i) You borrow money for a project

ii) You choose an easy, quick solution now instead of a better approach that takes longer

iii) The client doesn't pay

iv) The hardware is too old

Correct Answer: ii) You choose an easy, quick solution now instead of a better approach that takes longer

44. **Which is the best practice for variable naming?**

i) Use single letters like a, b, c

ii) Use descriptive and meaningful names

iii) Use random numbers

iv) Use as many characters as possible

Correct Answer: ii) Use descriptive and meaningful names

45. **Integrated Development Environment (IDE) is used to:**

i) Print documents

ii) Provide comprehensive facilities to programmers for software development

iii) Manage payroll

iv) Replace the developer

Correct Answer: ii) Provide comprehensive facilities to programmers for software development

46. **Black-box testing focuses on:**

i) Internal code logic

ii) Functional requirements and inputs/outputs

iii) Variable names

iv) Loop structures

Correct Answer: ii) Functional requirements and inputs/outputs

47. **White-box testing focuses on:**

i) System external behavior

ii) Internal paths, structures, and implementation

iii) User experience

iv) Documentation quality

Correct Answer: ii) Internal paths, structures, and implementation

48. **Unit testing is performed on:**

i) The entire system

ii) Individual modules or components

iii) The user manual

iv) The hardware

Correct Answer: ii) Individual modules or components

49. **Regression testing is done to:**

i) Test new features only

ii) Ensure that new changes haven't broken existing functionality

iii) Check the spelling

iv) Test the database speed

Correct Answer: ii) Ensure that new changes haven't broken existing functionality

50. **Alpha testing is performed by:**

i) The end-users at their own site

ii) The developers/testers at the developer's site

iii) The government

iv) Competitors

Correct Answer: ii) The developers/testers at the developer's site

51. **Corrective maintenance is done to:**

i) Add new features

ii) Fix bugs found after delivery

iii) Improve performance

iv) Change the UI color

Correct Answer: ii) Fix bugs found after delivery

52. **Adaptive maintenance involves:**

i) Fixing errors

ii) Modifying software to work in a new environment (e.g., new OS)

iii) Rewriting the entire code

iv) Deleting old files

Correct Answer: ii) Modifying software to work in a new environment (e.g., new OS)

53. **Perfective maintenance aims to:**

i) Fix critical crashes

ii) Enhance existing functions and improve performance

iii) Move to a new database

iv) Update the user list

Correct Answer: ii) Enhance existing functions and improve performance

54. **Software evolution refers to:**

i) The software becoming slower over time

ii) The process of continuous change throughout the software lifecycle

iii) Software being replaced by AI

iv) The death of a project

Correct Answer: ii) The process of continuous change throughout the software lifecycle

55. **The "Law of Increasing Complexity" states that:**

i) Software becomes simpler as it evolves

ii) As an evolving program is continually changed, its complexity increases unless work is done to maintain it

iii) Only hardware is complex

iv) Complex software is always better

Correct Answer: ii) As an evolving program is continually changed, its complexity increases unless work is done to maintain it

56. **Quality Assurance is _____ oriented.**

i) Product

ii) Process

iii) Error

iv) Management

Correct Answer: ii) Process

57. **ISO 9126 is a standard for:**

i) Hardware safety

ii) Software quality characteristics

iii) Network protocols

iv) Database design

Correct Answer: ii) Software quality characteristics

58. **Verification answers the question:**

i) Are we building the right product?

ii) Are we building the product right?

iii) Is the product expensive?

iv) Does the client like it?

Correct Answer: ii) Are we building the product right?

59. **Validation answers the question:**

i) Are we building the product right?

ii) Are we building the right product?

iii) Is the code fast?

iv) Is the documentation complete?

Correct Answer: ii) Are we building the right product?

60. **A "Walkthrough" is a type of:**

i) Dynamic testing

ii) Static analysis/Review

iii) Performance test

iv) Beta test

Correct Answer: ii) Static analysis/Review

61. **Lines of Code (LOC) is a measure of:**

i) Software quality

ii) Software size

iii) Software security

iv) User satisfaction

Correct Answer: ii) Software size

62. **Function Point (FP) analysis is used to:**

i) Measure the functionality provided to the user

ii) Count the number of brackets in code

iii) Measure CPU speed

iv) Calculate the electricity used

Correct Answer: i) Measure the functionality provided to the user

63. **Cyclomatic complexity measures:**

i) The number of lines

ii) The logical complexity of a program

iii) The number of users

iv) The cost of the project

Correct Answer: ii) The logical complexity of a program

64. **MTBF stands for:**

i) Mean Time Between Failures

ii) Max Time Before Finish

iii) Mode Time Binary File

iv) Main Task Build Frame

Correct Answer: i) Mean Time Between Failures

65. **Defect Density is calculated as:**

i) Total code / Total bugs

ii) Number of defects / Size of software

iii) Time / Defects

iv) Cost / Defects

Correct Answer: ii) Number of defects / Size of software

66. **SCM is responsible for:**

i) Fixing hardware

ii) Controlling changes in software

iii) Writing user manuals

iv) Marketing the product

Correct Answer: ii) Controlling changes in software

67. **A "Baseline" in SCM is:**

i) The bottom of the screen

ii) A specification or product that has been formally reviewed and agreed upon

iii) The first line of code

iv) The minimum price of software

Correct Answer: ii) A specification or product that has been formally reviewed and agreed upon

68. **Version Control systems (like Git) allow:**

i) Only one person to work at a time

ii) Tracking changes and collaborating on code

iii) Deleting history permanently

iv) Speeding up the internet

Correct Answer: ii) Tracking changes and collaborating on code

69. **"Branching" in version control refers to:**

i) Deleting the project

ii) Creating a separate line of development

iii) Joining two companies

iv) Writing comments

Correct Answer: ii) Creating a separate line of development

70. **Configuration Audit is used to ensure:**

i) The PC is fast enough

ii) The software meets defined requirements and SCM procedures

iii) The developer is working hard

iv) The office is clean

Correct Answer: ii) The software meets defined requirements and SCM procedures

71. **Reactive risk management strategy means:**

i) Planning for risks before they happen

ii) "Firefighting" or dealing with risks when they occur

iii) Ignoring risks entirely

iv) Hiring a risk manager

Correct Answer: ii) "Firefighting" or dealing with risks when they occur

72. **A "Risk" in software engineering is:**

i) A definite failure

ii) A potential future event that may lead to project failure

iii) A past mistake

iv) A successful feature

Correct Answer: ii) A potential future event that may lead to project failure

73. **Risk Analysis involves:**

i) Risk estimation and risk projection

ii) Only writing code

iii) Buying insurance

iv) Asking for more money

Correct Answer: i) Risk estimation and risk projection

74. **RMMM plan stands for:**

i) Risk Management, Mitigation, and Monitoring

ii) Real Management Modality Method

iii) Risk Money Main Mode

iv) Rapid Movement Maintenance Model

Correct Answer: i) Risk Management, Mitigation, and Monitoring

75. **Which is a technical risk?**

i) Loss of a key developer

ii) Ambiguous requirements

iii) Change in government policy

iv) Market competition

Correct Answer: ii) Ambiguous requirements

76. **The "CIA Triad" in security stands for:** i) Central Intelligence Agency

ii) Confidentiality, Integrity, Availability

iii) Coding, Implementation, Analysis

iv) Control, Interest, Authority

Correct Answer: ii) Confidentiality, Integrity, Availability

77. **Encryption is used to ensure:**

i) Availability

ii) Confidentiality

iii) Cost reduction

iv) Speed

Correct Answer: ii) Confidentiality

78. **A "Vulnerability" is:**

i) A security patch

ii) A weakness in the system that can be exploited

iii) A strong password

iv) A backup file

Correct Answer: ii) A weakness in the system that can be exploited

79. **SQL Injection is a type of:**

i) Hardware failure

ii) Software security attack

iii) Project management style

iv) Testing tool

Correct Answer: ii) Software security attack

80. **Principle of Least Privilege means:**

i) Everyone should have admin access

ii) Users should only have the minimum access necessary to do their job

iii) No one should have access

iv) Security is not important

Correct Answer: ii) Users should only have the minimum access necessary to do their job

81. **What is the primary goal of Software Engineering?**

i) To produce high-quality software within budget and time

ii) To make computers faster

iii) To reduce the number of developers

iv) To create beautiful icons

Correct Answer: i) To produce high-quality software within budget and time

82. **Evolutionary process models are:**

i) Linear

ii) Iterative

iii) Static

iv) One-time

Correct Answer: ii) Iterative

83. **In Scrum, what is a "User Story"?**

i) A technical specification

ii) A short description of a feature from the perspective of a user

iii) A bug report

iv) A manager's biography

Correct Answer: ii) A short description of a feature from the perspective of a user

84. **Data flow diagrams (DFD) show:**

i) How data moves through a system

ii) Database schema

iii) Hardware layout

iv) Project timeline

Correct Answer: i) How data moves through a system

85. **A project manager's primary focus is:**

i) Coding only

ii) Managing resources, time, and risks

iii) Cleaning the server room

iv) Designing logos

Correct Answer: ii) Managing resources, time, and risks

86. **Object-oriented design focuses on:**

i) Functions

ii) Objects and classes

iii) Binary code

iv) Paperwork

Correct Answer: ii) Objects and classes

87. **UI "Usability" refers to:**

i) How many colors are used

ii) How easy and efficient a system is to use

iii) The price of the software

iv) The size of the install file

Correct Answer: ii) How easy and efficient a system is to use

88. **The "Pipe and Filter" architecture is common in:**

i) Operating systems (e.g., Unix)

ii) Simple websites

iii) Mobile games

iv) Word processors

Correct Answer: i) Operating systems (e.g., Unix)

89. **What is "Spaghetti Code"?**

i) Well-organized code

ii) Unstructured and difficult-to-maintain code

iii) Code written in Italy

iv) A new programming language

Correct Answer: ii) Unstructured and difficult-to-maintain code

90. **Boundary Value Analysis (BVA) is a:**

i) White-box technique

ii) Black-box technique

iii) Management style

iv) Security protocol

Correct Answer: ii) Black-box technique

91. **"Threat Modeling" is done during:**

i) Coding

ii) Design and Planning

iii) Deployment

iv) Retirement

Correct Answer: ii) Design and Planning

92. **Which model is a "Risk-driven" process model?**

i) Waterfall

ii) Spiral

iii) V-Model

iv) Linear

Correct Answer: ii) Spiral

93. **Agile development prefers:**

i) Face-to-face communication

ii) Formal emails only

iii) No communication

iv) Paper-based memos

Correct Answer: i) Face-to-face communication

94. **A "Bug" is formally known as:**

i) An enhancement

ii) A fault or defect

iii) A feature

iv) Documentation

Correct Answer: ii) A fault or defect

95. **Beta testing is done by:**

i) Developers

ii) Real users in their own environment

iii) Robots

iv) Managers

Correct Answer: ii) Real users in their own environment

96. **The main purpose of SCM is to ensure:**

i) Integrity of the software product

ii) Speed of the internet

iii) Beauty of the code

iv) Low salary for developers

Correct Answer: i) Integrity of the software product

97. **The "Lehman's Laws" are associated with:**

i) Software design

ii) Software evolution

iii) Hardware manufacturing

iv) Project bidding

Correct Answer: ii) Software evolution

98. **Which is the most expensive type of maintenance?**

i) Corrective

ii) Adaptive

iii) Perfective

iv) Emergency

Correct Answer: iii) Perfective

99. **Software re-engineering is the process of:**

i) Deleting the system

ii) Examination and alteration of a system to reconstitute it in a new form

iii) Changing the project manager

iv) Buying new hardware

Correct Answer: ii) Examination and alteration of a system to reconstitute it in a new form

100. **"Legacy Systems" are:**

i) New systems

ii) Old systems that are still in use and business-critical

iii) Systems with no bugs

iv) Systems that have been deleted

Correct Answer: ii) Old systems that are still in use and business-critical

101. **CMMI stands for:**

i) Capability Maturity Model Integration

ii) Computer Management Model Info

iii) Coding Maturity Method Internal

iv) Central Model Main Integration

Correct Answer: i) Capability Maturity Model Integration

102. **Level 5 of CMMI is called:**

i) Defined

ii) Managed

iii) Optimizing

iv) Initial

Correct Answer: iii) Optimizing

103. **A "Software Audit" is performed by:**

i) The lead developer

ii) An independent body or team

iii) The client's family

iv) The HR department

Correct Answer: ii) An independent body or team

104. **Six Sigma focuses on:**

i) Reducing variability and defects

ii) Increasing the number of developers

iii) Using six different languages

iv) Writing 600 lines of code daily

Correct Answer: i) Reducing variability and defects

105. **Pareto Principle (80/20 rule) in SQA suggests:**

i) 80% of bugs are in 20% of the code

ii) 80% of developers are lazy

iii) 20% of the project is testing

iv) 80% of the cost is hardware

Correct Answer: i) 80% of bugs are in 20% of the code

106. **Which metric measures the time it takes to fix a bug?**

i) MTBF

ii) MTTR (Mean Time To Repair)

iii) LOC

iv) Function Point

Correct Answer: ii) MTTR (Mean Time To Repair)

107. **Halstead's Software Science metrics are based on:**

i) Number of operators and operands

ii) Number of pages in documentation

iii) Number of meetings

iv) Cost of the office

Correct Answer: i) Number of operators and operands

108. **"Fan-in" of a module refers to:**

i) The number of modules that call it

ii) The number of modules it calls

iii) The number of lines it contains

iv) The number of users

Correct Answer: i) The number of modules that call it

109. **"Fan-out" of a module refers to:**

i) The number of modules it calls

ii) The number of modules that call it

iii) The speed of the module

iv) The memory used

Correct Answer: i) The number of modules it calls

110. **A "Product Metric" measures:**

i) The development process

ii) The software product itself (e.g., size, complexity)

iii) The salary of the team

iv) The marketing budget

Correct Answer: ii) The software product itself (e.g., size, complexity)

111. **The "Check-out" operation in SCM means:**

i) Finishing the project

ii) Getting a copy of a file from the repository to work on

iii) Paying the developers

iv) Deleting a file

Correct Answer: ii) Getting a copy of a file from the repository to work on

112. **The "Check-in" (or Commit) operation means:**

i) Arriving at the office

ii) Saving changes back to the repository

iii) Logging out of the system

iv) Buying new software

Correct Answer: ii) Saving changes back to the repository

113. **What is a "Conflict" in version control?**

i) Two managers arguing

ii) When two people change the same line of code in different ways

iii) When the computer crashes

iv) When the internet is slow

Correct Answer: ii) When two people change the same line of code in different ways

114. **A "Repository" is:**

i) A place where all versions of files are stored

ii) A trash can

iii) A printer

iv) A meeting room

Correct Answer: i) A place where all versions of files are stored

115. **SCCB stands for:**

i) Software Configuration Control Board

ii) System Coding Control Body

iii) Software Change Central Bureau

iv) Simple Code Control Base

Correct Answer: i) Software Configuration Control Board

116. **Risk Mitigation means:**

i) Ignoring the risk

ii) Taking steps to reduce the impact or probability of a risk

iii) Making the risk happen faster

iv) Blaming the client

Correct Answer: ii) Taking steps to reduce the impact or probability of a risk

117. **Risk Projection is also known as:**

i) Risk Estimation

ii) Risk Coding

iii) Risk Deletion

iv) Risk Documentation

Correct Answer: i) Risk Estimation

118. **A "Known Risk" is one that:**

i) Can be uncovered after careful evaluation

ii) Is impossible to predict

iii) Has already happened

iv) Only the manager knows

Correct Answer: i) Can be uncovered after careful evaluation

119. **Risk Monitoring involves:**

i) Checking if risks are still present and if mitigation is working

ii) Watching movies about risks

iii) Hiring a security guard

iv) Increasing the project price

Correct Answer: i) Checking if risks are still present and if mitigation is working

120. **Which is a business risk?**

i) Building an excellent product that no one wants

ii) A bug in the login page

iii) A slow database

iv) A missing semicolon

Correct Answer: i) Building an excellent product that no one wants

121. **The "A" in CIA Triad stands for:**

i) Authentication

ii) Availability

iii) Accuracy

iv) Authorization

Correct Answer: ii) Availability

122. **What is "Social Engineering" in security?**

i) Writing code with friends

ii) Manipulating people into giving up confidential information

iii) Building a social network

iv) Designing a park

Correct Answer: ii) Manipulating people into giving up confidential information

123. **Cross-Site Scripting (XSS) is a:**

i) Web security vulnerability

ii) Desktop error

iii) Printer problem

iv) Project management tool

Correct Answer: i) Web security vulnerability

124. **Multi-Factor Authentication (MFA) requires:**

i) Only a password

ii) Two or more pieces of evidence to verify identity

iii) A large team

iv) High-speed internet

Correct Answer: ii) Two or more pieces of evidence to verify identity

125. **"Patching" software is done to:**

i) Add more bugs

ii) Fix security vulnerabilities and errors

iii) Increase the price

iv) Delete the database

Correct Answer: ii) Fix security vulnerabilities and errors

126. **What does UML stand for?**

i) Unified Modeling Language

ii) Universal Management Level

iii) Unique Model Logic

iv) Unified Maker List

Correct Answer: i) Unified Modeling Language

127. **A "Class Diagram" shows:**

i) The flow of data

ii) Static structure of classes and their relationships

iii) Time spent on a project

iv) Hardware connections

Correct Answer: ii) Static structure of classes and their relationships

128. **Software reliability is defined as:**

i) The probability of failure-free operation for a specific time

ii) The number of lines of code

iii) The beauty of the UI

iv) The speed of the developer

Correct Answer: i) The probability of failure-free operation for a specific time

129. **Which is an example of an Agile framework?**

i) Waterfall

ii) Scrum

iii) V-Model

iv) COCOMO

Correct Answer: ii) Scrum

130. **What is a "Stakeholder"?**

i) Only the developer

ii) Anyone who has an interest in or is affected by the project

iii) A piece of hardware

iv) A type of code comment

Correct Answer: ii) Anyone who has an interest in or is affected by the project

131. **Formal methods use _____ to specify software.**

i) Simple English

ii) Mathematical notation

iii) Hand drawings

iv) Video clips

Correct Answer: ii) Mathematical notation

132. **Reverse Engineering is used to:**

i) Create code from a design

ii) Recover design or requirements from existing code

iii) Delete the code

iv) Sell the code

Correct Answer: ii) Recover design or requirements from existing code

133. **The primary goal of a "Sprint Review" is to:**

i) Review the budget

ii) Demonstrate the work done to stakeholders

iii) Fire a team member

iv) Change the project name

Correct Answer: ii) Demonstrate the work done to stakeholders

134. **"Brooks' Law" states that:**

i) Adding manpower to a late project makes it later

ii) More developers make the project faster

iii) Coding is easy

iv) Testing is unnecessary

Correct Answer: i) Adding manpower to a late project makes it later

135. **Software "Portability" refers to:**

i) How heavy the computer is

ii) Ease with which software can be transferred between environments

iii) The number of images used

iv) The length of the manual

Correct Answer: ii) Ease with which software can be transferred between environments

136. **Unit testing is usually done by:**

i) The client

ii) The developers themselves

iii) An external auditor

iv) The marketing team

Correct Answer: ii) The developers themselves

137. **A "Data Dictionary" contains:**

i) Definitions of all data objects used in the system

ii) A list of employees

iii) Phone numbers of clients

iv) Computer brands

Correct Answer: i) Definitions of all data objects used in the system

138. **"Refactoring" does NOT:**

i) Improve code readability

ii) Change the observable behavior of the code

iii) Reduce complexity

iv) Remove duplication

Correct Answer: ii) Change the observable behavior of the code

139. **The most common model for calculating software cost is:**

i) COCOMO

ii) Waterfall

iii) Agile

iv) Spiral

Correct Answer: i) COCOMO

140. **Which diagram shows the sequence of messages between objects?**

i) Use Case Diagram

ii) Sequence Diagram

iii) State Diagram

iv) Class Diagram

Correct Answer: ii) Sequence Diagram

141. **Software "Interoperability" is:**

i) Ability of two or more systems to exchange information

ii) Ability to run on one computer only

iii) Ability to delete itself

iv) Ability to hide bugs

Correct Answer: i) Ability of two or more systems to exchange information

142. **In the Spiral Model, "Quadrant 1" is for:**

i) Risk analysis

ii) Planning and objective setting

iii) Engineering/Coding

iv) Evaluation

Correct Answer: ii) Planning and objective setting

143. **What is a "Smoke Test"?**

i) Testing if the computer catches fire

ii) A quick test to ensure the most crucial functions work

iii) A test for the printer

iv) A test for high-speed internet

Correct Answer: ii) A quick test to ensure the most crucial functions work

144. **Which testing checks if the system can handle a heavy load?**

i) Stress Testing

ii) Unit Testing

iii) Alpha Testing

iv) Logic Testing

Correct Answer: i) Stress Testing

145. **Which model is also called the "Classic Life Cycle"?**

i) Waterfall Model

ii) Agile Model

iii) RAD Model

iv) Prototype Model

Correct Answer: i) Waterfall Model

146. **What is "Feasibility Study"?**

i) Checking if the project is technically and financially possible

ii) Writing the final report

iii) Hiring new staff

iv) Selling the software

Correct Answer: i) Checking if the project is technically and financially possible

147. **Requirement "Elicitation" means:**

i) Deleting requirements

ii) Gathering requirements from stakeholders

iii) Testing requirements

iv) Coding requirements

Correct Answer: ii) Gathering requirements from stakeholders

148. **A "State Chart" diagram shows:**

i) The budget state

ii) The different states an object can have and transitions between them

iii) The location of the office

iv) The names of developers

Correct Answer: ii) The different states an object can have and transitions between them

149. **Integration testing focuses on:**

i) Individual modules

ii) Interactions between modules

iii) The entire system environment

iv) The user's mood

Correct Answer: ii) Interactions between modules

150. **What is "Walkthrough"?**

i) Walking in a park

ii) A peer review where the author leads the team through a document

iii) A formal government audit

iv) A type of coding language

Correct Answer: ii) A peer review where the author leads the team through a document

151. **Which tool is used for SCM?**

i) Git

ii) Photoshop

iii) MS Excel

iv) VLC Player

Correct Answer: i) Git

152. **A "Heuristic Evaluation" is used in:**

i) Database design

ii) User Interface (UI) testing

iii) Risk management

iv) Cost estimation

Correct Answer: ii) User Interface (UI) testing

153. **Software "Usability" includes:**

i) Learnability, Efficiency, Memorability, Errors, and Satisfaction

ii) Coding speed

iii) Internet speed

iv) Hardware price

Correct Answer: i) Learnability, Efficiency, Memorability, Errors, and Satisfaction

154. **The "Mainline" in version control is:**

i) The primary branch of code (e.g., master or main)

ii) The power cord

iii) The first line of a function

iv) The manager's desk

Correct Answer: i) The primary branch of code (e.g., master or main)

155. **"Validation" ensures that:**

i) The product meets user needs

ii) The code has no semicolons

iii) The developer is on time

iv) The computer is new

Correct Answer: i) The product meets user needs

156. **Extreme Programming (XP) emphasizes:**

i) Long planning phases

ii) Small releases and continuous feedback

iii) No testing

iv) Large teams

Correct Answer: ii) Small releases and continuous feedback

157. **What is a "Hotfix"?**

i) A bug that makes the PC warm

ii) A quick fix for a critical issue in production

iii) A new feature

iv) A type of coffee

Correct Answer: ii) A quick fix for a critical issue in production

158. **Software "Maturity" in CMMI refers to:**

i) The age of the software

ii) The extent to which a specific process is explicitly defined and managed

iii) The age of the developer

iv) The number of pages in the manual

Correct Answer: ii) The extent to which a specific process is explicitly defined and managed

159. **Which is NOT a software metric?**

i) Lines of Code

ii) Cyclomatic Complexity

iii) Function Points

iv) Keyboard weight

Correct Answer: iv) Keyboard weight

160. **"Dark Mode" in UI design is an example of:**

i) Requirement Engineering

ii) User Preference/Accessibility

iii) Software Risk

iv) SCM

Correct Answer: ii) User Preference/Accessibility

161. **What is "End-of-Life" (EOL) for software?**

i) When the software is first released

ii) When the software is no longer supported or sold

iii) When the computer breaks

iv) When the developer retires

Correct Answer: ii) When the software is no longer supported or sold

162. **Software "Efficiency" is related to:**

i) Amount of computing resources and code used

ii) Number of people in the office

iii) The font size

iv) The color of the mouse

Correct Answer: i) Amount of computing resources and code used

163. **A "Bug Bash" is:**

i) A party for bugs

ii) An event where many people test the software to find bugs

iii) A type of virus

iv) A programming contest

Correct Answer: ii) An event where many people test the software to find bugs

164. **"Dead Code" is:**

i) Code that is never executed

ii) Code that deletes files

iii) Code for a funeral home

iv) Code written in a dead language

Correct Answer: i) Code that is never executed

165. **"Sandboxing" in security is:**

i) Playing with sand

ii) Running a program in an isolated environment

iii) Deleting files

iv) Writing code on a beach

Correct Answer: ii) Running a program in an isolated environment

166. **What is a "Deliverable"?**

i) A pizza

ii) A tangible work product produced during the project

iii) A phone call

iv) A hardware part

Correct Answer: ii) A tangible work product produced during the project

167. **A "Milestone" is:**

i) A specific point in time used to measure project progress

ii) A heavy rock

iii) A type of code

iv) A distance marker

Correct Answer: i) A specific point in time used to measure project progress

168. **"Modularization" helps in:**

i) Making the system more complex

ii) Easier maintenance and understanding

iii) Reducing the number of features

iv) Making the file size larger

Correct Answer: ii) Easier maintenance and understanding

169. **"Static Analysis" is done:**

i) By running the code

ii) Without executing the code

iii) By the user

iv) By the client

Correct Answer: ii) Without executing the code

170. **Which is a functional requirement?**

i) The system must be fast

ii) The system must allow users to reset passwords

iii) The system must be secure

iv) The system must be green

Correct Answer: ii) The system must allow users to reset passwords

171. **Which is a non-functional requirement?**

i) The system must process 1000 transactions per second

ii) The system must generate invoices

iii) The system must save data

iv) The system must send emails

Correct Answer: i) The system must process 1000 transactions per second

172. **A "Data Flow Diagram" level 0 is also called:**

i) Context Diagram

ii) Detailed Diagram

iii) Code Diagram

iv) Risk Diagram

Correct Answer: i) Context Diagram

173. **"Encapsulation" is a principle of:**

i) Waterfall model

ii) Object-Oriented Programming/Design

iii) Risk Management

iv) UI Design

Correct Answer: ii) Object-Oriented Programming/Design

174. **A "Scrum Board" is used to:**

i) Play games

ii) Visualize tasks and their progress

iii) Write the SRS

iv) Calculate the budget

Correct Answer: ii) Visualize tasks and their progress

175. **What is "Burn-down Chart"?**

i) A chart showing remaining work vs time

ii) A chart showing office temperature

iii) A list of deleted files

iv) A chart of fire exits

Correct Answer: i) A chart showing remaining work vs time

176. **In the V-Model, "System Testing" is linked to:**

i) Coding

ii) Requirements Specification

iii) High-level Design

iv) Low-level Design

Correct Answer: iii) High-level Design

177. **What is "Equivalence Partitioning"?**

i) Dividing code into equal parts

ii) A black-box testing technique that divides input data into classes

iii) Sharing the budget equally

iv) Splitting the team

Correct Answer: ii) A black-box testing technique that divides input data into classes

178. **"Defect Leakage" happens when:**

i) Bugs are found by the end-user after release

ii) The developer finds a bug

iii) The computer leaks oil

iv) The manual is missing

Correct Answer: i) Bugs are found by the end-user after release

179. **"Monkey Testing" refers to**

i) Testing by monkeys

ii) Randomly providing inputs to the system to see if it crashes

iii) Testing for animal websites

iv) Using a banana-shaped mouse

Correct Answer: ii) Randomly providing inputs to the system to see if it crashes

180. **A "Logic Bomb" is:**

i) A piece of code that triggers a malicious action when certain conditions are met

ii) A type of hardware

iii) A loud noise from the CPU

iv) A mathematical puzzle

Correct Answer: i) A piece of code that triggers a malicious action when certain conditions are met

181. **"Agile" development is:**

i) Incremental and Iterative

ii) Linear and Sequential

iii) Static and Fixed

iv) Slow and Expensive

Correct Answer: i) Incremental and Iterative

182. **The "Product Backlog" contains:**

i) A list of all desired features for the product

ii) A list of all previous employees

iii) The office address

iv) The hardware specs

Correct Answer: i) A list of all desired features for the product

183. **A "Spike" in Agile is:**

i) A sharp object

ii) A task aimed at research or prototyping

iii) A sudden increase in price

iv) A bug

Correct Answer: ii) A task aimed at research or prototyping

184. **"TDD" stands for:**

i) Test Driven Development

ii) Total Design Document

iii) Time During Development

iv) Technical Data Design

Correct Answer: i) Test Driven Development

185. **"Clean Code" is:**

i) Code that has been washed

ii) Code that is easy to read and maintain

iii) Code with no comments

iv) Code written in a clean room

Correct Answer: ii) Code that is easy to read and maintain

186. **A "Null Pointer" error occurs in:**

i) UI design

ii) Implementation/Coding

iii) Risk management

iv) Marketing

Correct Answer: ii) Implementation/Coding

187. **"Cross-platform" software can:**

i) Only run on Windows

ii) Run on multiple operating systems

iii) Only run on one phone

iv) Be used by one person only

Correct Answer: ii) Run on multiple operating systems

188. **"Load Testing" checks:**

i) How heavy the laptop is

ii) System behavior under expected user load

iii) If the electricity is on

iv) The size of the box

Correct Answer: ii) System behavior under expected user load

189. **"Unit" in unit testing refers to:**

i) A whole building

ii) The smallest testable part of an application

iii) One dollar

iv) One developer

Correct Answer: ii) The smallest testable part of an application

190. **"Pair Programming" involves:**

i) Two programmers working on two different projects

ii) Two programmers working together at one workstation

iii) Two computers connected by a cable

iv) Two managers in one meeting

Correct Answer: ii) Two programmers working together at one workstation

191. **What is "Technical Debt"?**

i) Money owed to a bank

ii) The implied cost of future reworking caused by choosing an easy solution now

iii) A slow computer

iv) A missing keyboard

Correct Answer: ii) The implied cost of future reworking caused by choosing an easy solution now

192. **"UAT" stands for:**

i) User Acceptance Testing

ii) Universal Access Tool

iii) Unit Analysis Template

iv) User Architecture Task

Correct Answer: i) User Acceptance Testing

193. **A "Fishbone Diagram" (Ishikawa) is used for:**

i) Drawing fish

ii) Identifying root causes of a problem

iii) Designing a database

iv) Calculating speed

Correct Answer: ii) Identifying root causes of a problem

194. **The "Waterfall Model" is:**

i) Sequential

ii) Iterative

iii) Circular

iv) Random

Correct Answer: i) Sequential

195. **"Software Configuration" includes:**

i) Code, documentation, and data

ii) Only the EXE file

iii) Only the user manual

iv) Only the desk and chair

Correct Answer: i) Code, documentation, and data

196. **"Risk Exposure" is calculated as:**

i) Probability * Impact

ii) Probability + Impact

iii) Time / Cost

iv) Size * 10

Correct Answer: i) Probability * Impact

197. **"Authentication" is:**

i) Giving access rights

ii) Verifying who a user is

iii) Hiding data

iv) Deleting users

Correct Answer: ii) Verifying who a user is

198. **"Authorization" is:**

i) Verifying who a user is

ii) Determining what a user can do

iii) Writing a book

iv) Selling a license

Correct Answer: ii) Determining what a user can do

199. **"Sanitization" of input helps prevent:**

i) SQL Injection

ii) Hardware rust

iii) Dust in the office

iv) Keyboard errors

Correct Answer: i) SQL Injection

200. **The ultimate goal of software engineering is to produce:**

i) Any software

ii) High-quality, reliable, and maintainable software

iii) Complex code

iv) Documentation only

Correct Answer: ii) High-quality, reliable, and maintainable software